

1. Evaluate $\sum_{k \in S} \frac{1}{k^3}$, where S is the set of positive integers such that $\frac{1}{k}$ has a terminating decimal expansion.
2. Show that every positive integer is a sum of one or more numbers of the form $2^r 3^s$, where r and s are nonnegative integers and no summand divides the other. For example, $23 = 9 + 8 + 6$.
3. Find all positive integers less than 10000 which are palindromic in both base 10 and base 11.
4. A chord of constant length slides around in a semicircle. The midpoint of the chord and the projections of its ends upon the base form the vertices of a triangle. Prove that the triangle is isoscles and never changes its shape.
5. Prove that for any positive real numbers a, b, c ,

$$\frac{a^2}{b} + \frac{b^2}{c} + \frac{c^2}{a} + \sqrt{ab} + \sqrt{bc} + \sqrt{ca} \geq 2(a + b + c).$$

6. Find all triples (p, q, r) of primes such that

$$p^2 + 2q^2 + r^2 = 3pqr.$$